

EEAIC assignment4

Alessandro Vegni & Giulio Guerrieri

Part 1 - Boost Converter

Parameter	Value	Parameter	Value
V_{IN}	1.0 V	R_{ind} (ESR L)	40 m Ω
V_{OUT}	2.0 V	C	1 μ F
I_{OUT}	0.5 A	f_{SW}	10 MHz
L	470 nH	R_{DSON}	80 m Ω

1. Circuit in Phase 1 and Phase 2

During Phase 1 ($0 < t < DT_s$): the inductor L is charged by V_{IN} , while the capacitor C supplies the load R_{load} . During Phase 2 ($DT_s < t < T_s$): the inductor discharges its energy into the output capacitor and load.

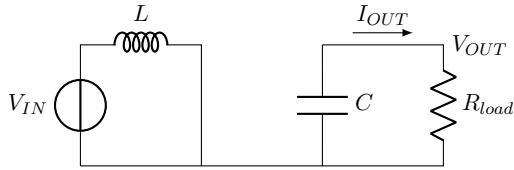


Figure 1: Phase 1 (S_1 ON, S_2 OFF)

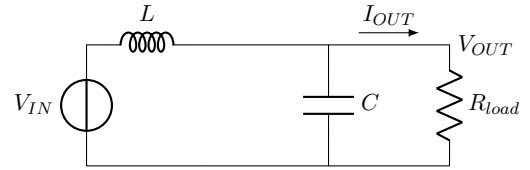
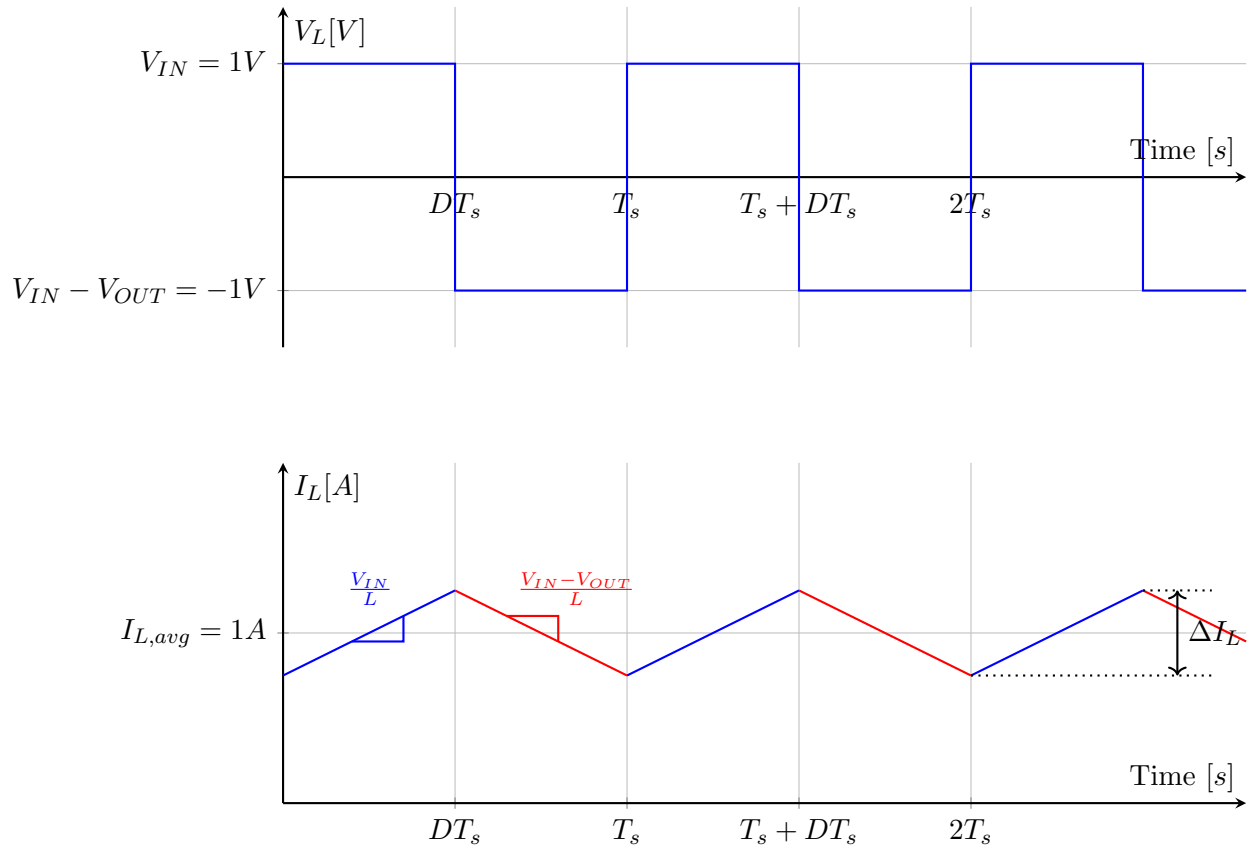


Figure 2: Phase 2 (S_1 OFF, S_2 ON)

2. Inductor Voltage and Current Waveforms

Applying Kirchhoff's Voltage Law, the inductor voltage V_L is positive (V_{IN}) during Phase 1, causing the inductor current I_L to ramp up. In Phase 2, V_L becomes negative ($V_{IN} - V_{OUT}$), causing I_L to ramp down. Under the small-ripple approximation, these linear ramps form a triangular AC ripple superimposed on the DC average current.



3. Inductor Voltage Balance

In steady-state operation, the principle of inductor volt-second balance dictates that the average voltage across the inductor over one full switching period must be exactly zero. Equating the areas of the positive and negative voltage intervals yields the ideal DC voltage conversion ratio M .

$$\langle V_L \rangle = \frac{1}{T_s} \int_0^{T_s} V_L(t) dt = D(V_{IN}) + (1 - D)(V_{IN} - V_{OUT}) = 0$$

$$V_{IN} - (1 - D)V_{OUT} = 0 \implies M = \frac{V_{OUT}}{V_{IN}} = \frac{1}{1 - D}$$

Given $V_{IN} = 1.0V$ and $V_{OUT} = 2.0V$, $M = 2$. Therefore, $D = 0.5$.

4. Capacitor Current Waveform

By applying Kirchhoff's Current Law at the output node, we define the capacitor current I_C . In Phase 1, the disconnected inductor forces the capacitor to discharge entirely into the load, yielding a constant negative current. In Phase 2, the inductor replenishes the capacitor while simultaneously feeding the load.

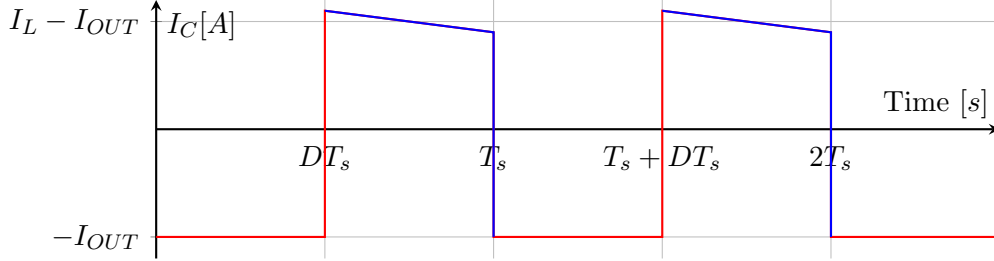


Figure 3: Capacitor Current Waveform.

5. Capacitor Charge Balance

By the principle of capacitor charge balance, the average current flowing through the output capacitor in steady-state must be zero. This constraint mathematically links the average inductor current to the output load current:

$$\langle I_C \rangle = \frac{1}{T_s} \int_0^{T_s} I_c(t) dt = D(-I_{OUT}) + (1-D)(I_{L,avg} - I_{OUT}) = 0$$

$$-I_{OUT} + (1-D)I_{L,avg} = 0 \implies I_{L,avg} = \frac{I_{OUT}}{1-D}$$

Given $I_{OUT} = 0.5A$ and $D = 0.5$, $\langle I_L \rangle = I_{L,avg} = 1.0A$.

6. Inductor Current Slope

Using the fundamental inductor characteristic equation $V_L = L \frac{di_L}{dt}$, the slopes of the current waveforms are directly proportional to the applied phase voltages.

Phase 1: $\frac{dI_L}{dt} = \frac{V_{IN}}{L} = 2.127 \times 10^6 \text{ A/s}$

Phase 2: $\frac{dI_L}{dt} = \frac{V_{IN}-V_{OUT}}{L} = -2.127 \times 10^6 \text{ A/s}$

7. Inductor Current Ripple

$$V_L = L \frac{\Delta I_L}{\Delta t} = L \frac{\Delta I_L}{DT_s}$$

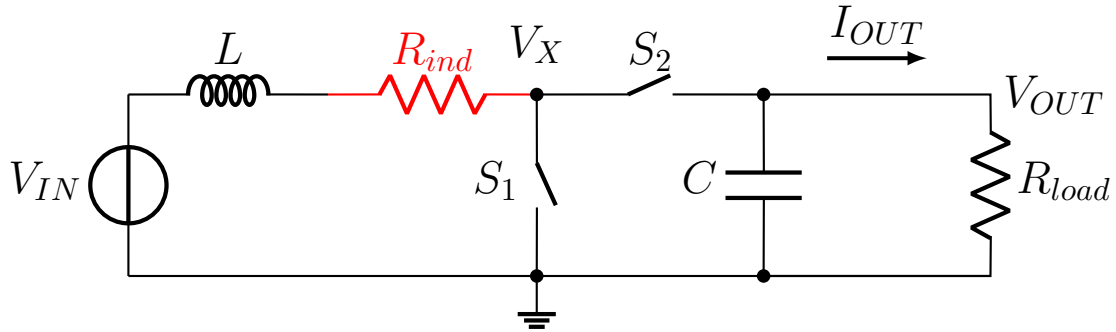
$$\Delta I_L = \frac{V_{IN}}{L} DT_s = 106.38 \text{ mA}$$

8. V_{OUT} Voltage Ripple

$$\Delta V_{OUT} = \frac{\Delta Q}{C} = \frac{I_{OUT} \cdot DT_s}{C} = 0.025 \text{ V} = 25 \text{ mV}$$

9. Reformulate M with Inductor Series Resistance R_{ind}

Real-world inductors possess a parasitic equivalent series resistance (ESR), denoted as R_{ind} . This resistance introduces a voltage drop proportional to the inductor current. When this drop is factored into the volt-second balance equation, it reveals that the actual voltage conversion ratio diverges from the ideal, particularly at high duty cycles.



Using the modified voltage balance equation:

$$\langle V_L \rangle = D(V_{IN} - I_L R_{ind}) + (1 - D)(V_{IN} - I_L R_{ind} - V_{OUT}) = 0$$

Substituting $I_L = \frac{V_{OUT}}{R_{load}} \frac{1}{(1-D)}$ yields:

$$V_{IN} - \frac{V_{OUT}}{(1-D)} \frac{R_{ind}}{R_{load}} - (1-D)V_{OUT} = 0 \implies M = \frac{V_{OUT}}{V_{IN}} = \frac{1-D}{(1-D)^2 + \frac{R_{ind}}{R_{load}}}$$

10. Plot M vs D

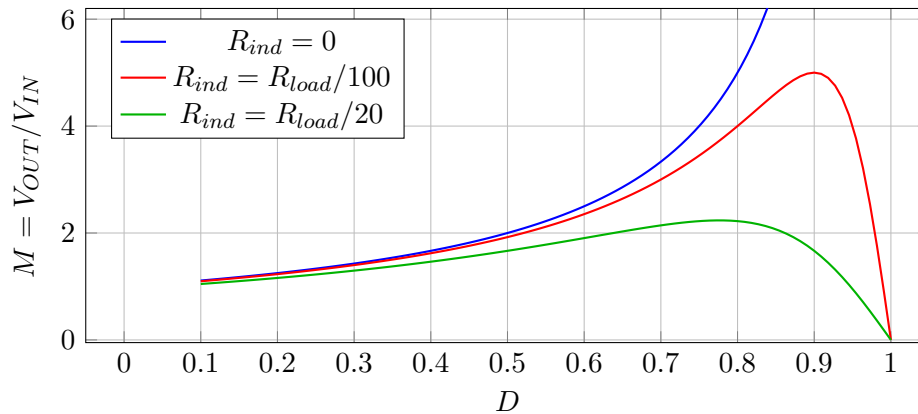


Figure 4: Voltage Conversion Ratio M vs Duty Cycle D for different R_{ind} values.

Figure 4 plots the non-ideal conversion ratio M against the duty cycle D. As expected, parasitic elements progressively degrade the conversion ratio.

11. Efficiency vs M including R_{ind}

$$\begin{aligned}\eta &= \frac{P_{OUT}}{P_{IN}} = \frac{(V_{OUT})^2/R_{load}}{V_{IN}I_{IN}} \\ &= \frac{(V_{OUT})^2/R_{load}}{V_{OUT} \frac{1-D}{(1-D)^2 + \frac{R_{ind}}{R_{load}}} \frac{V_{OUT}}{R_{load}} \frac{1}{(1-D)}} \\ &= \frac{(1-D)^2}{(1-D)^2 + \frac{R_{ind}}{R_{load}}} = (1-D)M\end{aligned}$$

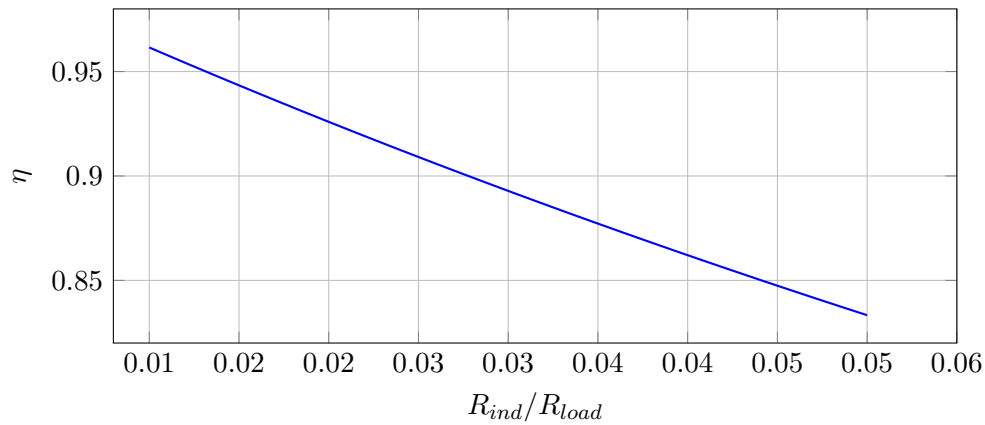
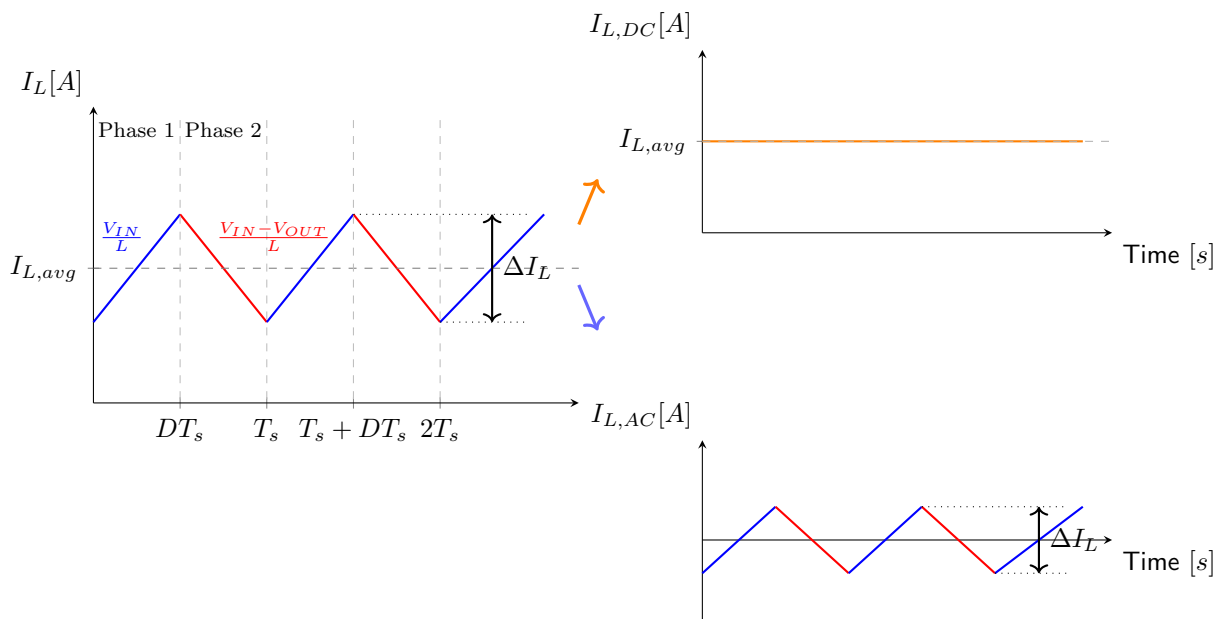


Figure 5: Efficiency η vs normalized inductor resistance R_{ind}/R_{load} for $D = 0.5$.

12. Inductor DC and AC Conduction Losses



The inductor current can be split in 2 contributions, DC and AC.

The AC current contribution has a triangular shape, and we know that its RMS value will be

its peak divided by $\sqrt{3}$.

Thus, the inductor loss can be estimated as follows: $P_L = I_{RMS,L}^2 \cdot R_{ind} \approx 40 \text{ mW}$

$$\begin{aligned} I_{RMS,L}^2 &= I_{RMS,DC}^2 + I_{RMS,AC}^2 = \left(\frac{I_{load}}{(1-D)} \right)^2 + \left(\frac{I_{peak}}{\sqrt{3}} \right)^2 = \\ &= (2I_{load})^2 + \left(\frac{\Delta I_L \cdot 0.5}{\sqrt{3}} \right)^2 = 4I_{load}^2 + \frac{\Delta I_L^2}{12} \end{aligned}$$

It is interesting to note that despite the different topologies, the power loss in the Boost converter configuration yields a result identical to that of a Buck converter; indeed, the RMS current flowing through the inductor remains the same as $I_{load,Boost} = 0.5 \cdot I_{load,Buck}$.

13. S1 and S2 Conduction Losses P_{cond}

Conduction loss refers to the power dissipated as heat when current flows through the internal resistance of a switch while it is in the "ON" state.

$$P_{cond,S1} = D \cdot I_{L,rms}^2 \cdot R_{DS(on)} \approx 40.04 \text{ mW}$$

$$P_{cond,S2} = (1-D) \cdot I_{L,rms}^2 \cdot R_{DS(on)} \approx 40.04 \text{ mW}$$

Total $P_{cond} = 80.08 \text{ mW}$.

14. S1 and S2 Switching Losses P_{sw}

As the transistor gate capacitance will be charged and discharged every cycle, its switching loss can be estimated as follows:

$$P_{sw,S1} = P_{sw,S2} = C_{gate} f_{sw} V_{gate,swing}^2 = 16.2 \text{ mW}$$

Total $P_{sw} = 32.4 \text{ mW}$.

15. Converter Efficiency considering P_{cond} only

Assuming nominal $P_{OUT} = 1.0 \text{ W}$:

$$\eta_{cond} = \frac{P_{OUT}}{P_{OUT} + P_L + P_{cond,S1} + P_{cond,S2}} \approx 89.28\%$$

16. Converter Efficiency considering P_{cond} and P_{sw}

$$\eta_{total} = \frac{P_{OUT}}{P_{OUT} + P_L + P_{cond} + P_{sw}} \approx 86.77\%$$

Part II - Simulation on Cadence

1. Ideal Switch TestBench Screenshots

By setting up the simulation according to our circuit parameters ($R_{ind} = 1 \text{ m}\Omega$, $R_{DS(on)} = 1 \text{ m}\Omega$), we obtained the following outputs, which are fully consistent with the theoretical predictions derived previously.

The plotted waveforms (V_L , I_L , V_{OUT} , I_C) and the extracted steady-state metrics (ΔI_L , ΔV_{OUT} , $\bar{V}_{OUT}$, $\bar{I}_L$) are reported in the figures below.

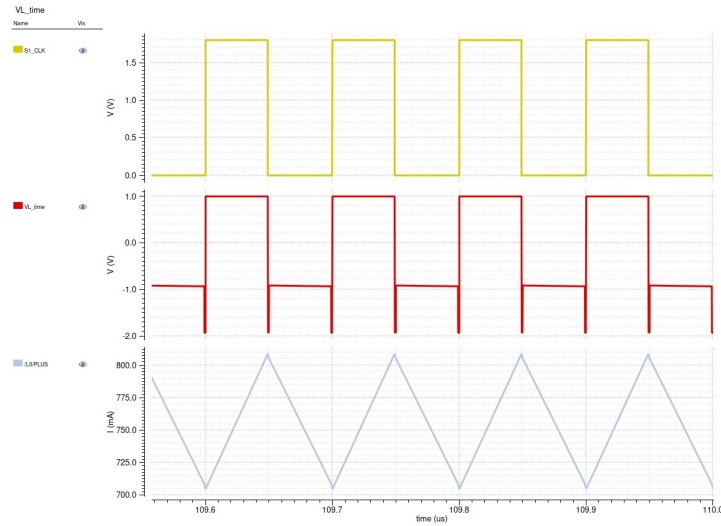


Figure 6: V_L , I_L with respect to the bottom Switch clock

As observed in the V_L waveform, there are prominent negative peaks reaching approximately -2 V during the non-overlapping time, specifically right before the bottom switch closes. This phenomenon occurs because both switches are momentarily OFF (dead time), but the inductor current must maintain its continuous flow. Consequently, the inductor pulls the switching node voltage very high enough to forward-bias the diode of the bottom switch, creating a temporary conduction path, which is immediately resolved as soon as the bottom switch actively turns on and shorts the node to ground.

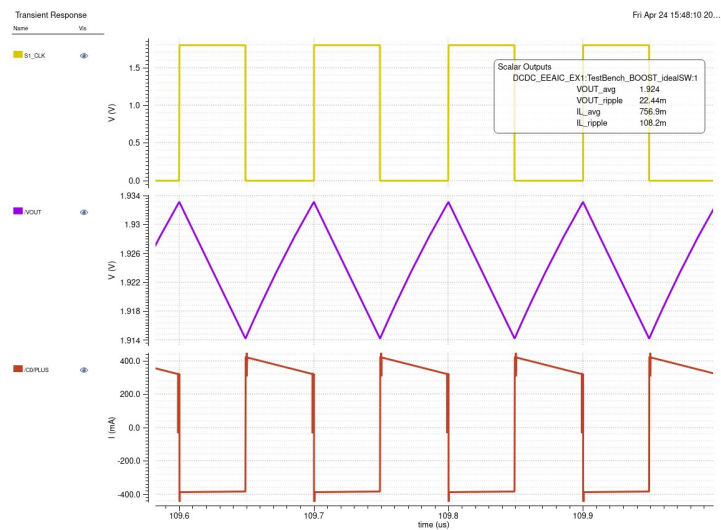


Figure 7: V_{OUT} , I_C with respect to the bottom Switch clock

Additionally, the I_C waveform exhibits sharp current spikes during the non-overlapping times, driven by the parasitic elements of the switch models. However, these transient peaks are practically negligible in terms of overall energy and circuit operation, as their duration is extremely short (approximately 2 ps).

2. Plot M vs D

Moving forward, the impact of parasitic components on the voltage conversion ratio M was evaluated. The duty cycle D was swept from 0.2 to 0.9, keeping $V_{IN} = 1.0$ V and $R_{load} = 4 \Omega$.

The following plot compares the resulting $M(D)$ curves for the three requested scenarios: nearly ideal conditions (a), introduction of inductor losses (b), and the combined effect of both inductor and switch ON-resistances (c). As expected, the curves highlight how parasitic elements progressively degrade the conversion ratio, especially at higher duty cycles.

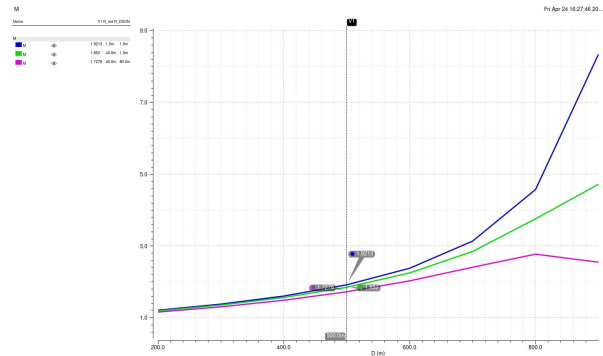


Figure 8: $M(D)$ for different R_{ind} and $R_{DS(on)}$

3. Plot Efficiency vs D

Following the analysis of the voltage conversion ratio, the converter's efficiency (η) was evaluated under the exact same operating conditions and parameter sweeps.

The plot below illustrates the efficiency versus duty cycle for the three previously defined scenarios. As expected, the nearly ideal case (a) maintains a high efficiency across the sweep. However, the introduction of inductor losses (b) and switch ON-resistances (c) progressively degrades the performance. This degradation becomes particularly severe at higher duty cycles, where the increased average currents lead to dominant I^2R conduction losses.

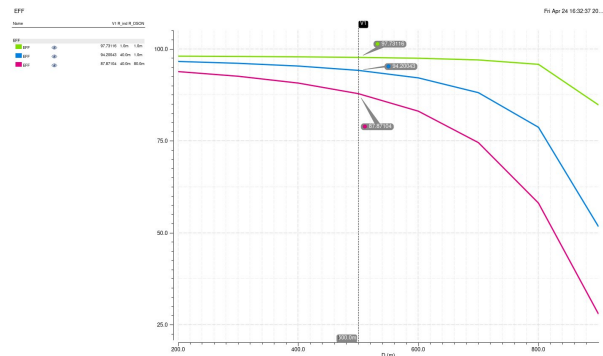


Figure 9: $\eta(D)$ for different R_{ind} and $R_{DS(on)}$

4. Transistor Sizing for $R_{DS(on)} = 80m\Omega$

To size the S_1 and S_2 switches using the `nch_25` NMOS transistors from the `tsmcN65` library, a dedicated testbench was built to characterize the device's ON-resistance. The schematic of this setup is presented in Figure 10.

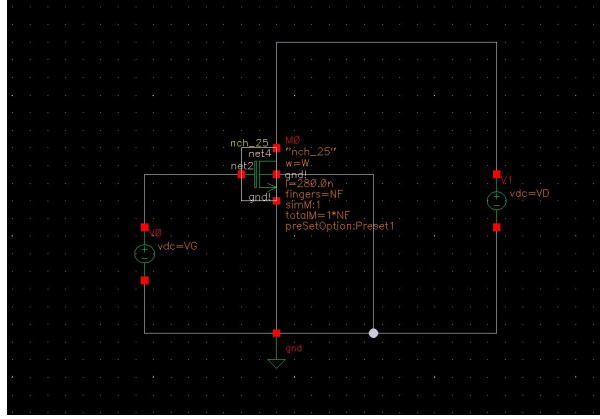


Figure 10: *nch 25* testbench setup

Keeping the channel length fixed at the technology minimum, a parametric sweep of the transistor width (W) was performed to target the desired resistance of 80 m Ω . The simulation sweep yielded, as shown in Figure 11, a total width of:

$$W_{80m\Omega, R_{dson}} = 286\mu\text{m} \times 50 = 14.3\text{mm}$$

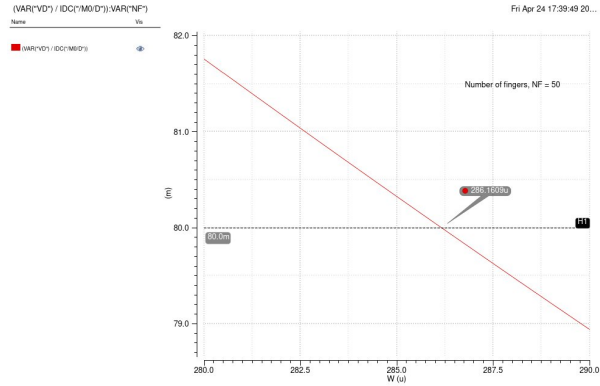


Figure 11: $R_{DS(on)}(W)$

It must be noted that such a massive dimension is highly impractical for a physical layout in a typical 65 nm technology node.

5. Efficiency Maximization Sizing with only conduction losses

As suggested, the sizing process was carried out using the provided `TestBench_BOOST_realSW` schematic with $R_L = 1\text{m}\Omega$ for both simulations. For the sake of simplicity the following architectural choice is taken in the tb: an NMOS transistor is used for the high-side switch. In a boost converter, fully turning on a high-side NMOS requires a gate voltage ($V_{G2} = 3.7\text{ V}$ instead of $V_{G1} = 1.8\text{ V}$) significantly higher than the targeted output voltage ($V_{OUT} = 2.0\text{ V}$). This presupposes the availability of an external higher voltage reference or a dedicated bootstrap circuit, which is an unusual and more complex approach compared to a standard PMOS implementation for low-voltage applications.

To maximize the efficiency considering only conduction losses, a parametric sweep was performed on the device sizing for both S_1 and S_2 . The base transistor width was fixed at $100\ \mu\text{m}$, and the multiplier parameter (m) was swept.

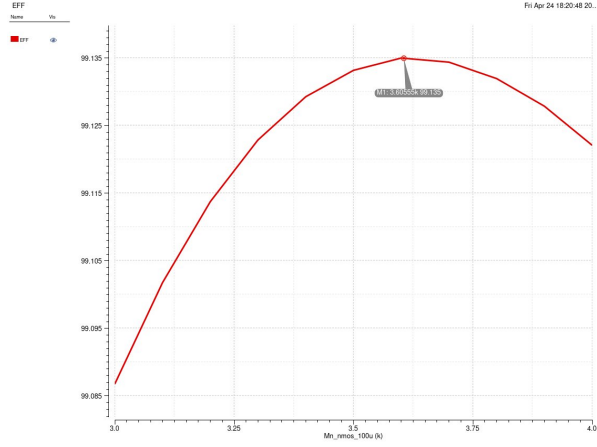


Figure 12: $\eta(M_{nmos\ 100\mu\text{m}\ mult})$

The simulation results in Figure 12 indicate that the maximum efficiency is theoretically achieved with a multiplier $M_{nmos\ 100\mu\text{m}\ mult} = 3.6k$. While this mathematical optimum successfully minimizes the $R_{DS(on)}$ and the associated conduction losses, it results in a massive total effective width of over 300 mm.

6. Efficiency Maximization Sizing with conduction and switching losses

In this final stage of the sizing process, the efficiency analysis was refined by including switching losses, specifically the power dissipated to charge and discharge the gate capacitances (P_{gate}). Unlike the previous case, where only the converter's input power was considered, the total efficiency η is now calculated by accounting for the power delivered by the gate drivers:

$$\eta_{total} = \frac{P_{OUT}}{P_{OUT} + P_{cond} + P_{gate\ drivers}} \quad (1)$$

A parametric sweep of the multiplier ($M_{nmos\ 100\mu\text{m}\ mult}$) was performed to identify the new optimal operating point. The resulting efficiency curve, shown in Figure 13, exhibits an optimum where the total efficiency peaks at 96.58%. This is a significant decrease compared to the 99.135% obtained in the previous analysis, as the power budget now accounts for the energy required to drive the transistor gates.

The optimal efficiency is now achieved at a slightly lower multiplier value of $M_{nmos\ 100\mu\text{m}\ mult} = 3k$. This behavior perfectly illustrates the design trade-off in power converters: while increasing the multiplier continues to reduce conduction losses (I^2R), it simultaneously causes a linear increase in P_{gate} due to the larger total gate charge (Q_g). This result confirms that in high-frequency switching converters, the penalty of driving large parasitic capacitances eventually outweighs the benefit of a lower ON-resistance, setting a physical and practical limit to the device size.

Note: The increased roughness in the efficiency curve, compared to the conduction-only analysis, is attributed to the inclusion of high-frequency switching transients. While conduction losses depend on relatively stable signals, switching losses involve integrating extremely narrow current spikes. Small variations in the simulator's numerical time-stepping across the parametric sweep lead to minor fluctuations in the integral of these spikes, resulting in the observed numerical noise.

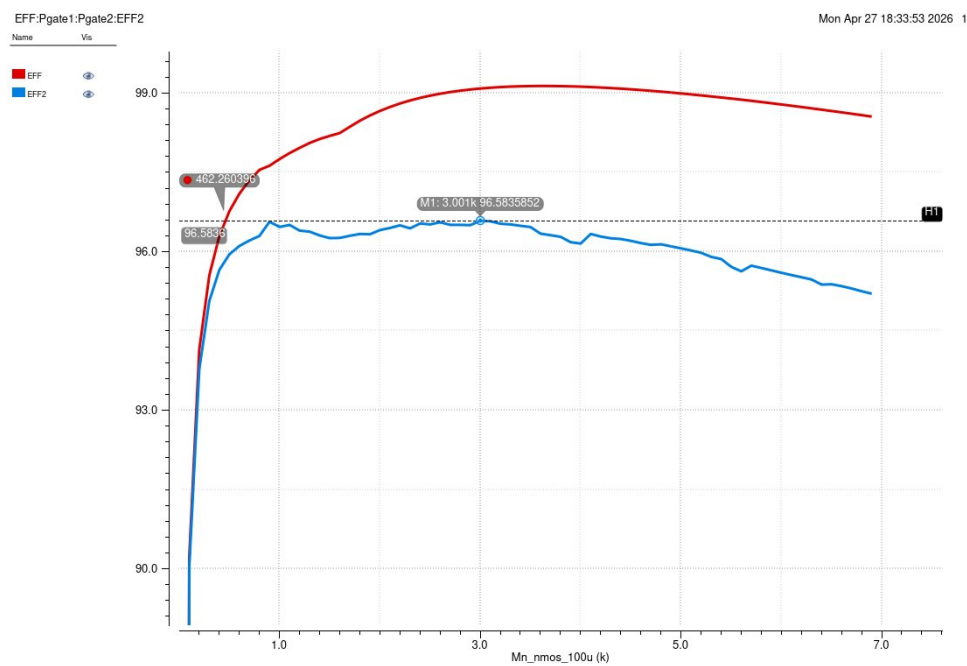


Figure 13: Total efficiency vs multiplier M , showing the trade-off between conduction and switching losses.

Energy-efficient Analog Integrated Circuits

DCDC Converters – Exercise 1

Alessandro Novello

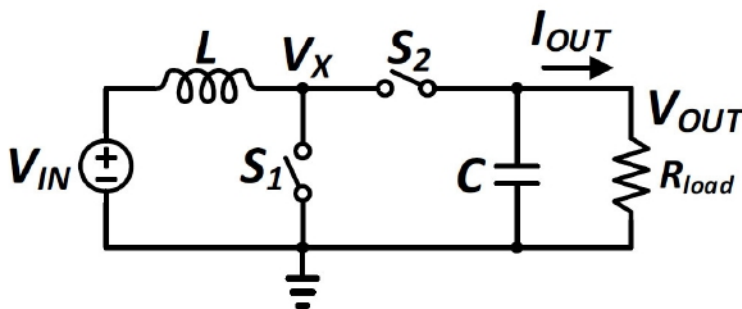
Prof. Taekwang Jang

Introduction

In this assignment you will analyze and design a Boost converter following the modeling you have learnt in the Tutorial. You can find the testbench files in `/home/eeaic_01fs26/DCDC_EEAIC_EX1`, you can copy the files to your local directory.

Part I – Modeling of the Boost Converter

Given a standard Boost converter topology and its specifications in the table:



V _{IN}	1.0V
V _{OUT}	2.0V
I _{OUT}	0.5A
L	470nH
RL (ESR_L)	40mOhm
C	1uF
f _{sw}	10MHz
R _{DS(on)}	80mOhm

1. **(1 Pt)** Draw circuit in phase 1 and phase 2
2. **(3 Pt)** Draw the inductor voltage and current waveforms in phase 1 and phase 2
3. **(3 Pt)** Use inductor voltage balance to determine $M=V_{OUT}/V_{IN}$
4. **(6 Pt)** Formulate and draw the capacitor C current waveform
5. **(3 Pt)** Use capacitor charge balance to determine the inductor average current
6. **(3 Pt)** Use inductor equation to calculate the inductor current slope
7. **(3 Pt)** Calculate inductor current ripple

8. (6 Pt) Calculate the V_{OUT} voltage ripple
9. (6 Pt) Include the inductor series resistance R_{ind} and reformulate M
10. (6 Pt) Plot M vs D in the 3 following cases:
 - a. $R_{load}=0.66$; $R_{ind}=0$;
 - b. $R_{load}=0.66$; $R_{ind}=R_{load}/100$;
 - c. $R_{load}=0.66$; $R_{ind}=R_{load}/20$;
 - i. For all three cases, please, use the following settings:
 1. $D=0.1:0.001:1$;
 2. $xlim([-0.05 \ 1.05])$
 3. $ylim([-0.05 \ 6.2])$
11. (6 Pt) Calculate the efficiency vs M including R_{ind} for $R_{load}=0.666$, $R_{ind}=R_{load} \cdot [0.01:0.001:0.05]$ and $D=0.5$
12. (3 Pt) Calculate and plot the inductor DC and AC conduction losses
13. (2 Pt) Calculate the S1 and S2 conduction losses P_{cond}
14. (3 Pt) Calculate the S1 and S2 switching losses P_{sw} accounting for their gate capacitance only, equal to $C_{gate}=500pF$ and $V_{gate,swing}=1.8V$
15. (3 Pt) Estimate the converter efficiency considering only P_{cond}
16. (3 Pt) Estimate the converter efficiency considering P_{cond} and P_{sw}

Part II – Simulation of the Boost Converter on Cadence

1. (10 Pt) Run the *TestBench_BOOST_idealSW* with $R_{ind}=1m\Omega$ and $R_{DSOn}=1m\Omega$, then include in your report the screenshots of the following variables plotted and show whether they match the previous calculation:
 - a. V_L , I_L , V_{OUT} , I_C , I_{L_ripple} , V_{OUT_ripple} , $V_{OUT_average}$, $I_{L_average}$
 - i. Make sure the y-axis is well readable and that the waveforms are clearly visible for all the signals
2. (5 Pt) Plot M vs $D=[0.2:0.1:0.9]$ for $V_{IN}=1.0V$ and $R_{load}=4$ in the following cases:
 - a. $R_L=1m$, $R_{DSOn}=1m$
 - b. $R_L=40m$, $R_{DSOn}=1m$
 - c. $R_L=40m$, $R_{DSOn}=80m$
3. (5 Pt) Plot Efficiency vs $D=[0.2:0.1:0.9]$ for $V_{IN}=1.0V$ and $R_{load}=4\Omega$ in the following cases:
 - a. $R_L=1m$, $R_{DSOn}=1m$
 - b. $R_L=40m$, $R_{DSOn}=1m$
 - c. $R_L=40m$, $R_{DSOn}=80m$
4. (5 Pt) Use *nch_25* NMOS type transistors from the *tsmcN65* library for S1 and S2 and size them to achieve $R_{DSOn}=80m\Omega$ using minimum length transistors, what is the width you obtained?

For simplicity of the gate driving and high-side design, please, use the schematic below:

TestBench_BOOST_realSW/schematic

5. (5 Pt) Size S1 and S2 and size them to maximize the efficiency @ $V_{OUT}/V_{IN}=2.0V/1.0V$ considering the conduction losses only
6. (10 Pt) Repeat 5 considering S1 and S2 switching losses as well.